

Christy Cheung

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EDUCATION

Cornell University, Ithaca, NY

Aug. 2024 – Dec. 2027

Bachelor of Science in Computer Science, Minor in AI

Relevant Courses: Object-Oriented Programming, Data Structures, Machine Learning, Computer Systems

EXPERIENCE

Amazon Web Services (AWS)

Jun. 2026 – Aug. 2026

Software Development Engineer Intern

- **Re-architected EC2 Spot capacity prediction service** to remove 3 redundant SageMaker endpoints and models across 37 AWS regions, reducing **request volume by 66%** and projected GPU inference **costs by \$150K** per year
- Implemented a containerized prediction service using **caching and AWS S3** to reuse capacity forecasts across 3+ applications, eliminating redundant computation and \$40K in annual infrastructure overhead
- **Improved the EC2 Spot capacity prediction algorithm** for rebalance notifications, achieving 98.2% accuracy against production capacity and extending customer notice time before interruptions from **2 minutes to 1 hour**
- **Eliminated a single point of failure** in the prediction pipeline by introducing partition-level error isolation, preventing individual timeouts from triggering full-region retries across ~10K capacity predictions

VortexNet

Dec. 2025 – Jan. 2026

Software Engineering Intern

- Delivered an end-to-end **full-stack** audio-to-text **web application** using **ML integration** to streamline addition of audio transcriptions and video subtitles for 50+ languages in collaboration with product and ML teammates
- Introduced **JWT-secured REST APIs** with FastAPI and **Celery + Redis async job orchestration** and in-memory **caches** to offload ML inference from API services, maintaining P90 request latency under 500 ms
- Migrated from in-memory storage to **PostgreSQL relational databases** with SQLAlchemy and Alembic, eliminating data loss on restarts and ensuring data integrity across API services and background workers
- Deployed a containerized backend API using Docker and **AWS ECS** behind an **Application Load Balancer**, integrating **RDS, ElastiCache, and S3** for scalable cloud deployment and team-wide **CI/CD workflows**

KPMG

July 2025 – Aug. 2025

Machine Learning Intern

- **Automated sensitive information detection workflows** across 3 enterprise applications in collaboration with compliance teams, reducing manual compliance review time of 400+ documents by 60.3%
- **Integrated object detection into automation scripts** to identify 15+ sensitive stamp types on financial documents, achieving 93% detection accuracy and improving audit reliability of document processing pipelines
- Spearheaded adoption of **computer vision models** in cross-functional teams, reducing compute and memory usage

PROJECTS

Physical Caregiving Language Model | Website, Paper

Sep. 2025 – Feb. 2026

- Produced an AI workflow to **retrieve patient-specific clinical evidence** from 270,000+ medical publications to **inform user functionality estimation** and downstream assistive robot policy generation
- Integrated Retrieval-Augmented Generation pipeline to improve LLM's user evaluation by 37% on benchmark tasks
- **Published a multi-modal dataset** for assistive robots at the IEEE/RSJ IROS and a **benchmark for LLM** caregiving decision-making abilities in ACM/IEEE HRI, contributing 2 open resources for robotics research

AI Health Hackathon – Multilingual AI Triage Assistant

Mar. 2026 – Mar. 2026

- Architected a **multilingual voice AI assistant** using the **Gemini API** to monitor emergency room patient symptoms across 8 languages and support real-time triage workflows in healthcare
- Designed a **React** and Zustand dashboard integrated with the Gemini Live API to provide real-time patient updates and alerts, achieving 70% alert accuracy
- **Led a 6 member team** as project architect, owning end-to-end system design and technical coordination and winning the Cornell AI Health Tech Hackathon

TECHNICAL SKILLS

Programming: Python, Java, C, TypeScript/JavaScript, SQL, HTML, CSS, OCaml

Frameworks and Tools: React, FastAPI, Redis, Celery, Material UI, SQLAlchemy, Alembic, PyTorch

Cloud and DevOps: AWS (ECS, ECR, RDS, ElastiCache, CloudWatch, IAM), Docker, GitHub Actions, Git

Languages: Mandarin, Cantonese, English